

NEURAL BELLMAN OPERATORS

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We develop Neural Bellman Operators for economic control with recursive utility, endogenous preferences, and strategic interaction. Policy evaluation and feasible improvement are separated from independent economic verification. A differential implementation yields full-domain certificates for trained nonlinear consumption policies. A positive-weight implementation trains preference policies and dynamic-game actors with full finite-model best responses. We then train continuous preference controls without exhaustive maximizing labels and construct budget-safe enclosures over the complete action box. These enclosures remain valid when action search is unresolved and distinguish numerical policy loss from continuous-economy approximation error. Matched direct, classical, gated-network, and projection comparisons identify the costs of actors and local improvement. Common-noise payoff comparisons diagnose state-coverage failures in coupled nonquadratic economies. The results connect learned policies to explicit value and deviation accounts while retaining separate conclusions for continuous-domain bounds, nodal certificates, and sampled high-dimensional diagnostics.

KEYWORDS: Dynamic programming, Neural approximation, Recursive utility, Endogenous preferences, Viscosity solutions, Dynamic games.

1. INTRODUCTION

Quantitative economic models increasingly place several margins of adjustment inside the same dynamic decision problem. An investor chooses consumption and risk exposure while managing a preference state. A sophisticated consumer chooses today knowing that tomorrow's self will reconsider. A firm invests while anticipating the investment policies of its rivals. These problems share a computational structure: a continuation value must correctly evaluate a policy, and that policy must

satisfy the appropriate optimality or equilibrium condition. The two requirements are distinct. A perfectly evaluated policy need not be optimal, and an attractive policy evaluated by an inaccurate continuation value need not improve welfare.

This paper develops Neural Bellman Operators (NBO) around this distinction. A critic represents continuation utility. A feasible actor represents the controls that enter the economic generator. The critic is fitted to a policy-evaluation equation, including its terminal and boundary data. Holding the critic and its derivative channels fixed, the actor improves its own Hamiltonian. An independent maximization step measures the remaining feasible-action gap. The resulting error decomposition concerns economic values and deviations, rather than only parameter gradients or the scalar quantity minimized by an optimizer.

The motivation remains continuous-time economic control, including recursive utility and costly preference formation in the motivating model of [Qi \(2025\)](#). We do not replace these problems by a prediction task. Instead, we make explicit how a neural approximation is connected to the original decision problem. In the smooth case, the connection follows from a verification bound. In the nonsmooth case, it follows from a monotone Bellman approximation and a comparison principle. The actor and critic may be trained by different numerical procedures in either case. Their economic interpretation is determined by the equations they satisfy and by the errors that are verified after training.

The paper makes three contributions. First, it specifies a map for policy evaluation and improvement whose derivative graph, feasible policy class, boundary data, and verification output are explicit. Second, it provides an operational error account that connects differential and positive-weight implementations: a complete-domain differential certificate for nonlinear consumption, and a budget-safe complete-action enclosure for a two-state endogenous-preference economy. The latter remains valid with unresolved action boxes and requires neither exact maximizing training labels nor concavity of the interpolated control objective. Third, it implements and compares these constructions for learned preference policies, coupled capital economies, and dynamic strategic agents. Recursive utility and temporal selves remain within the same economic formulation. Comparison, monotonicity, and approximate policy iteration are inherited mathematical ingredients; the implemented maps, explicit

1 action-oracle distinction, and verified economic computations are the objects stud- 1
 2 ied here. 2

3 The numerical program uses generic multilayer critics and actors as well as clearly 3
 4 separated analytical laboratories. The finite preference and game studies train actors 4
 5 using exhaustively constructed continuation-payoff labels; their results establish 5
 6 neural policy representation, not avoided maximization. The continuous-action 6
 7 preference study instead differentiates the actor’s own payoff, records local im- 7
 8 provement as a hybrid policy, and reserves global action covering for independent 8
 9 verification after training. Direct optimization and an actor-free search ablation 9
 10 expose which parts of the computation produce the measured policy quality. Ad- 10
 11 ditional gated and projection comparisons exploit the same economic problem 11
 12 rather than weakening the baseline. Dense nonlinear capital experiments use paired 12
 13 pathwise payoffs to diagnose state coverage. Homothetic portfolios, temporal-self 13
 14 formulas, the nonsmooth exit problem, and the exact quadratic family remain dis- 14
 15 tinct economic and algebraic checks. No matrix calculation is counted as neural 15
 16 training, and no grid refinement difference is silently converted into a continuous- 16
 17 time bound. 17

1.1. *Relation to the literature*

21 Neural residual methods build on numerical dynamic programming and function 21
 22 approximation, not on a new principle of optimality. Projection and related methods 22
 23 exploit economic structure to approximate a Bellman equation; see [Judd \(1998\)](#). 23
 24 Derivative approximation by smooth neural networks provides useful represen- 24
 25 tation results ([Hornik et al., 1990](#)), but does not establish that a particular finite 25
 26 architecture or optimizer solves a nonlinear boundary-value problem. 26

27 The differential critic is closely related to the neural PDE approach in [Sirignano](#) 27
 28 [and Spiliopoulos \(2018\)](#). Their DGM formulation includes differential and boundary 28
 29 conditions and treats HJB equations. The deep BSDE method of [Han et al. \(2018\)](#) also 29
 30 includes an HJB application. The distinguishing object here is the combination of 30
 31 an explicit feasible actor, block-specific evaluation, and economic error accounting 31
 32 across control and equilibrium problems. An analytical formula for the maximizing 32

action is helpful when available, but is not a universal dividing line between neural PDE methods. Likewise, recursive dependence on the value is a nonlinear zeroth-order term; full nonlinearity in the PDE sense concerns the highest derivatives.

Recursive utility separates intertemporal substitution from risk aversion (Epstein and Zin, 1989, Duffie and Epstein, 1992). Its infinite-horizon formulation requires particular care about utility-process selection and transversality; Herdegen et al. (2021) explains why a stationary algebraic solution alone is insufficient. The time-inconsistency extension is an equilibrium problem in the sense of Björk et al. (2016), not an ordinary control problem with a relabelled flow payoff. Finally, stochastic approximation results require their own step-size, stability, and limiting-dynamics conditions (Borkar, 2024). They do not convert constant-step Adam into a convergence theorem for economic optimality.

Section 2 specifies the economy. Section 3 defines the implemented map and Section 4 establishes its error accounts. Section 5 completes the differential theorem–algorithm–experiment chain. The subsequent sections retain and extend the portfolio, recursive-utility, preference-adjustment, temporal-self, dynamic-game, and scaling applications. The appendix proves the principal results. The supplement gives arithmetic details, application-specific assumptions, every seed-level comparison, and sensitivity results.

2. THE CONTROLLED ECONOMY

Let W be an m -dimensional Brownian motion on a filtered probability space satisfying the usual conditions. The state $S = (u, X)$ contains internal and external state variables. An action $a = (\theta, c, \pi, \dots)$ combines internal adjustment and external choices. The controlled state evolves as

$$dS_s = b(s, S_s, a_s) ds + \sigma(s, S_s, a_s) dW_s, \quad a_s \in \mathcal{A}(s, S_s). \quad (1)$$

The compact feasible correspondence $\mathcal{A}$ is part of the model. Actions are progressively measurable and must generate a well-defined state process satisfying the declared state restrictions. State viability does not follow from bounded actor outputs alone.

We work initially on a finite horizon T . A computational domain D is either invariant, equipped with a specified reflection mechanism, or stopped on exit. In the stopped case let $\tau = T \wedge \inf\{s \geq t : S_s \notin D\}$. The payoff $g(\tau, S_\tau)$ is specified on the entire parabolic stopping boundary, including the terminal face. For a fixed admissible policy a , utility is the policy-specific solution

$$Y_s^a = g(\tau, S_\tau) + \int_s^\tau f(v, S_v, a_v, Y_v^a) dv - \int_s^\tau Z_v^a dW_v. \quad (2)$$

Adjustment costs are included in f . For example, time-additive utility has $f(s, S, a, y) = r_0(s, S, a) - \rho y$, where r_0 includes all current resource and adjustment costs. Define $V^a(t, x) = Y_t^{a, t, x}$ and $V^*(t, x) = \sup_a V^a(t, x)$. The value inside a candidate policy's recursive objective is its own utility process, not the unknown optimum inserted into every candidate objective.

ASSUMPTION 1—Finite-horizon comparison class: *The controlled state problem is well posed for the admissible policies used below. The coefficients are continuous, locally Lipschitz in the state uniformly over actions, and bounded on the compact stopped domain. The payoff is continuous and bounded. On an invariant utility interval containing all policy values, f is continuous in its arguments and uniformly Lipschitz in utility with constant $L \geq 0$. The admissible class is stable under concatenation and has the measurable-selection properties needed for dynamic programming. The resulting HJB problem and the fixed-policy problems obey comparison in the bounded continuous class with the specified boundary conditions.*

The comparison requirement identifies both a function class and a boundary problem. It is not asserted merely because the primitive coefficients are continuous. On a bounded, regular Dirichlet domain, a boundary barrier is one way to verify attainment of the stopping payoff. Reflection requires the corresponding Neumann or oblique derivative condition and a well-posed reflected state process. Unbounded-state applications require growth bounds and localization. These conditions are checked separately from a neural residual.

For a smooth function v , write

$$\mathcal{L}^a v = b(t, x, a)^\top Dv + \frac{1}{2} \text{tr}\{\sigma(t, x, a)\sigma(t, x, a)^\top D^2 v\}, \quad (3)$$

$$\mathcal{H}^a[v] = f(t, x, a, v) + \mathcal{L}^a v, \quad F[v] = -v_t - \sup_{a \in \mathcal{A}(t, x)} \mathcal{H}^a[v]. \quad (4)$$

The value solves $F[V^*] = 0$ with the specified boundary data, in the viscosity sense where classical derivatives do not exist. The diffusion covariance is the positive semidefinite matrix $\sigma\sigma^\top$. No claim that the product of two arbitrary positive semidefinite matrices is symmetric positive semidefinite is needed.

A continuous martingale representation can also be used when its quadratic variation is Markov in an augmented state. If $d\langle M \rangle_s = Q(s, S_s, a_s)ds$, the covariance in (3) becomes $\sigma Q\sigma^\top$. A merely predictable process Q_s is not, without further state augmentation, a deterministic coefficient of a Markov HJB equation. Quadratic variation by itself also does not imply conditionally Gaussian increments. The implementation in this revision uses the Brownian model (1).

2.1. Infinite horizon and economic normalization

For a stationary problem, the boundary-value equation is $\sup_a \{f(x, a, V) + \mathcal{L}^a V\} = 0$. We use an infinite-horizon result only after specifying admissibility, integrability, and the selected utility process. In the time-additive case, a sufficient verification requirement is that the localized Itô identities are uniformly integrable and $\mathbb{E}[e^{-\rho T} | V(S_T) |] \rightarrow 0$ for the policies being compared. Recursive problems require an appropriate utility comparison and transversality condition instead of simply borrowing the additive one.

Cardinal transformations also require care. Multiplying a fixed utility function by a positive constant does not change its maximizer. Adding a function of an *endogenously controlled preference state* generally does. Consumption units, preference normalization, and terminal utility are therefore primitive economic choices in Section 8.

3. NEURAL BELLMAN OPERATORS

The actor $a_\phi(t, x)$ and critic $v_\xi(t, x)$ represent policies and continuation values. An internal actor and an external actor can be separate parameter blocks; their joint action remains the input to the same economic generator. A smooth activation is

useful for exact differentiation, while a nonsmooth neural representation can be used with the positive-weight Bellman formulation below.

3.1. Feasibility and boundary data

For a box $[a, \bar{a}]$, a sigmoid map produces interior feasible actions but does not attain its endpoints at finite weights. Binding constraints can instead be represented by a projection onto the closed interval, by an explicit boundary candidate, or by a finite action set including the endpoints. The constrained portfolio calculation uses a projection and can attain a zero risky share exactly. On a simplex, a probability map supplies a feasible interior point and boundary faces must again be considered when certifying the optimum. General constraints require a feasible parameterization, a certified projection, or a constrained action solver. A finite penalty is not an exact feasibility certificate.

On a finite horizon, an elementary hard-terminal architecture is

$$v_{\xi}(t, x) = G(x) + (T - t)N_{\xi}(t, x). \quad (5)$$

This enforces the terminal face, but not automatically a lateral boundary. A compatible lifting G_B and a factor d_B vanishing on all Dirichlet faces give $v_{\xi} = G_B + d_B N_{\xi}$. Alternatively, boundary residuals are trained separately and retained explicitly in the error bound. A reflected preference state requires its own derivative condition; a terminal architecture does not enforce it. For recursive utility, a domain-preserving transformation must additionally keep the continuation value in the domain of the aggregator.

3.2. The block-specific computational graph

Let ν be a specified distribution of state–time points and let sg denote stop-gradient. A critic step minimizes

$$L_C(\xi; \bar{\phi}) = \mathbb{E}_{\nu} \left[-\partial_t v_{\xi} - \mathcal{H}^{\text{sg}(a_{\bar{\phi}})}[v_{\xi}] \right]^2 + L_B(\xi), \quad (6)$$

where L_B is absent only for boundary conditions enforced exactly. The policy, sampling locations, and any rival actions are detached. The value, time derivative, state gradient, and Hessian remain differentiable with respect to the critic parameters.

An actor step minimizes

$$L_A(\phi; \bar{\xi}) = -\mathbb{E}_v \mathcal{H}^{a\phi}[\text{sg}(v_{\bar{\xi}}), \text{sg}(Dv_{\bar{\xi}}), \text{sg}(D^2v_{\bar{\xi}})]. \quad (7)$$

All critic derivative channels are fixed in this step. There is no critic gradient from L_A and no actor gradient from L_C . A printed sum $L_A + \omega L_C$ may be a reporting device, but it is not jointly differentiated. The implementation exposes separate functions and regression tests for these derivative paths.

The distinction changes the population problem. With one action, $f(v) = -v$, $T = 1$, and terminal utility one, the correct solution is $v^*(t) = e^{t-1}$. The perturbation $v_\epsilon = v^* - \epsilon(1 - t)$ preserves the terminal condition. The joint objective used in the reviewed draft satisfies

$$\Delta(L_A + L_C) = -\frac{\epsilon}{2} + \frac{7\epsilon^2}{3}, \quad \Delta L_C = \frac{7\epsilon^2}{3}. \quad (8)$$

Thus the revised critic objective has the correct root as its minimizer along this perturbation, whereas the joint objective does not. The same problem is visible in a stationary equation: $-H + \omega H^2$ prefers $H = 1/(2\omega)$, not $H = 0$. No choice of finite positive weight resolves this identification error.

The actor objective is a local policy-improvement surrogate. Under differentiability of the policy value, the derivative of lifetime recursive utility from (t, x) instead involves the adjoint weight

$$D_\phi V^{a\phi}(t, x) = \mathbb{E}_{t,x} \int_t^\tau e^{\int_t^s f_y(v, S_v, a_v, V^{a\phi}) dv} D_a \mathcal{H}^{a\phi}[V^{a\phi}](s, S_s) D_\phi a_\phi(s, S_s) ds, \quad (9)$$

subject to the differentiability and integrability conditions given in the supplement. Arbitrary domain sampling does not reproduce this occupation and recursive-utility weighting. NBO uses (7) for approximate pointwise improvement and verifies the remaining action gap; it does not identify that loss with an exact lifetime policy gradient.

Algorithm 1 NBO evaluation, improvement, and acceptance

Require: Model, admissible actions, utility domain, boundary data, training and validation domains.

- 1: Initialize a feasible actor and a boundary-compatible critic.
- 2: **for** each outer iteration **do**
- 3: Freeze actor outputs. Update only the critic using (6) or a positive-weight Bellman evaluation residual.
- 4: Freeze the critic and every derivative channel. Improve only the actor using (7) or a feasible action solver.
- 5: Evaluate boundary error, policy-evaluation error, and the independent feasible-action gap.
- 6: Record the initialization, updates, conditioning, unsuccessful attempts, and stopping rule.
- 7: **if** the declared economic error criterion is met **then**
- 8: Return the policy, value, verification domain, and error report.
- 9: **end if**
- 10: **end for**
- 11: Return the computed approximation and its measured errors; do not infer optimality from optimizer stationarity.

3.3. Conditioning and stopping

A positive, parameter-independent state weighting can precondition the critic equation without changing its zero set. If that weight depends on learned quantities, its derivative must be included in the stated critic objective. The homothetic laboratory divides the Hamiltonian by its positive coefficient and differentiates that entire normalized critic residual. In the actor step the same coefficient is detached. Error conversion to the original economic equation includes the scale factor.

A zero residual target does not make learning dynamics stationary. For the scalar loss $\ell(z) = a^2 z^2 / 2$, gradient descent is stable only when $0 < \eta a^2 < 2$. Likewise, full differentiation of $\delta^2 / 2$ gives $\delta \nabla \delta$, while a semi-gradient TD update differentiates only the current-value term. The two computations must not be compared using

inconsistent signs or discount conventions. If f already includes $-\rho v$, a time discretization must not add a second independent discount to the same continuation equation.

Constant-step Adam is a numerical choice in the calibration laboratory, not an application of a two-timescale theorem. A genuine two-timescale analysis would require nonsummable and square-summable step sizes, an actor-to-critic ratio tending to zero, controlled noise, bounded iterates, and suitable attracting limiting dynamics (Borkar, 2024). A stationary critic gradient need not imply a zero residual, and a stationary multi-agent gradient need not imply Nash optimality. The results below therefore concern verified approximation errors rather than an assumed universal optimizer convergence property.

3.4. The implemented map and its verification output

Fix the economy, boundary realization, network classes, optimization budget, and verification procedure, denoted collectively by ϑ . Let $\mathcal{P}$ be the admissible feedback policies, Ξ the critic parameter space, and Ω the sampling streams. The implemented NBO is the numerical map

$$\mathcal{N}_\vartheta : \mathcal{P} \times \Xi \times \Omega \longrightarrow \mathcal{V}_\vartheta \times \mathcal{P}_\vartheta \times \mathcal{C}_\vartheta, \quad (a, \xi, \omega) \longmapsto (v, \tilde{a}, C). \quad (10)$$

The first component is approximate policy evaluation; the second is feasible improvement; the third records verification of the resulting pair. A record identifies the model, parameters, domain, boundary faces, evaluation error, maximization error, coverage, discretization errors, and unchecked assumptions. An unavailable bound is not zero. Failure to obtain a useful certificate is a possible output of the map. With its sampling stream fixed, the implementation is deterministic. We do not assert that a network learns an unrestricted Bellman operator on a function space.

The evaluation–improvement structure is an instance of approximate policy iteration, including its partial-evaluation variants (Scherrer et al., 2015). Neural representation does not introduce a new principle of optimality, and monotonicity belongs to the underlying positive-weight scheme, not to gradient descent. The

NBO specification joins the block-specific derivative graph to boundary and utility-domain restrictions and to a common economic error account. Sections 5 and 8.4 implement the differential and positive-weight versions of this map, respectively, with independent verification of their outputs.

3.5. Continuous-action proposals and a budget-safe Bellman enclosure

The learned map and its verifier have different computational roles. In the differential implementation, a feasible actor supplies an action and the critic evaluates that action's generator. In the positive-weight implementation, write

$$(T_n^a w)_i = r_{n,i}(a) + q_n \{P_{n,i}(a)w + b_{n,i}(a)\}, \quad (T_n w)_i = \sup_{a \in \mathcal{A}_i} (T_n^a w)_i. \quad (11)$$

Here $P_{n,i}(a)$ has nonnegative entries with sum at most one. The term b contains prescribed stopping payoffs, and $q_n > 0$. Continuous controls need not enter this expression concavely. Reflection, interpolation cells, and stopping decisions can change as the action changes. A supremum, rather than an attained maximizer, suffices below.

Our continuous-action implementation fits the actor by differentiating its own evaluated continuation payoff. It does not construct an all-action target or receive an exact maximizing label. A bounded tanh output enforces the control box. A local improvement step can subsequently compare that proposal with finitely many continuous perturbations. Such a step is recorded as part of a *hybrid* policy; it is not identified with the raw network and is not assumed globally optimal. Critic fitting uses the selected feasible action and the preceding continuation critic. Training ends before any exhaustive reference or action-cover verification begins.

For independent verification, freeze the complete policy $\pi = (\pi_n)$ and cover each action box by closed subboxes. At a state and time, let c_i be an outward lower bound obtained by evaluating any feasible action using a fixed continuation upper array. A box can be discarded only when its verified upper bound does not exceed $c_i + \tau$. Every box not so discarded is either subdivided or retained with its upper bound. Exhausting a time, depth, or box budget therefore enlarges the reported bound; it never converts an unresolved search into a success.

PROPOSITION 1—Budget-safe action and policy envelopes: Suppose the terminal arrays satisfy $L_N \leq g \leq U_N$. At each time n , evaluate the fixed-policy map outward to obtain $L_n \leq T_n^{\pi_n} L_{n+1}$. For every state, maintain a complete action cover whose box bounds enclose $T_n^a U_{n+1}$ for all actions in that box. Set $U_{n,i}$ to the maximum of the outward-rounded pruning threshold and all unresolved box upper bounds. Then, regardless of which boxes remain unresolved,

$$L_n \leq V_n^\pi \leq V_n^* \leq U_n, \quad 0 \leq V_n^* - V_n^\pi \leq U_n - L_n. \quad (12)$$

All inequalities are on the stated nodal economy. No concavity, exact training labels, or convergence of neural optimization is required. If $U_n - T_n U_{n+1} \leq e_n \mathbf{1}$, then its optimal-value excess is at most

$$\left(\prod_{j=n}^{N-1} q_j \right) \|U_N - g\|_\infty + \sum_{k=n}^{N-1} \left(\prod_{j=n}^{k-1} q_j \right) e_k.$$

The proof, including stopping faces, is in the supplement. A stored unresolved bracket supplies an upper bound for e_n ; it is not replaced by the requested tolerance. The policy lower envelope and the Bellman upper envelope are computed separately. The latter may be shared across several frozen policies on the same economy, while each policy requires its own evaluation. This separation prevents an expensive verification oracle from becoming undisclosed actor supervision.

In the preference application, the continuous box has three coordinates and the state grid has two. Bilinear interpolation is enclosed cell by cell. On each rectangle, its extrema occur at cell corners; a second implementation uses cell-slope enclosures. If a box crosses a wealth stopping face, the verifier takes the union of the interior continuation and the prescribed exit payoff. Their numerical values need not coincide at off-grid exit locations. Treating the entire action objective as globally differentiable at this switch would invalidate a derivative-only bound. Reflection is handled by including every crossed fold. Positive consumption and a preference interval strictly above one keep the powers and denominators in their stated domains.

The complete-action bound is not a continuous-state or continuous-time theorem. The verifier interprets stored primitive coefficients and nodal abscissae as exact

real inputs and encloses subsequent operations outward. Training and verification agree to numerical precision on the inherited action grid, but the certificate's precise object is this declared nodal economy, not an unspecified sequence of floating-point evaluations. The supplemental transfer result states separately the state interpolation, diffusion approximation, boundary monitoring, and action errors needed to recover the continuous economy. None is inferred from a sampled residual or silently assigned zero.

4. VERIFICATION, APPROXIMATION, AND VISCOSITY SELECTION

For a feasible policy a and a smooth candidate v , define

$$r^a[v] = -v_t - \mathcal{H}^a[v], \quad g^a[v] = \sup_{b \in \mathcal{A}} \mathcal{H}^b[v] - \mathcal{H}^a[v] \geq 0. \quad (13)$$

The first error concerns evaluation; the second concerns improvement. Their difference is the maximized HJB residual: $F[v] = r^a[v] - g^a[v]$. We keep the two quantities separate because the policy value and its regret depend differently on them.

THEOREM 2—Finite-horizon economic error bound: *Under Assumption 1, suppose a bounded smooth candidate v in the declared utility interval and an admissible Markov policy a satisfy $|r^a[v]| \leq e$, $0 \leq g^a[v] \leq \eta$ throughout the verification domain. Suppose the boundary payoff error is at most B . Any reflected boundary condition is imposed exactly. Set $A_L(s) = (e^{Ls} - 1)/L$ for $L > 0$ and $A_0(s) = s$. Then, for $s = T - t$,*

$$|v(t, x) - V^a(t, x)| \leq e^{Ls} B + A_L(s) e, \quad (14)$$

$$|v(t, x) - V^*(t, x)| \leq e^{Ls} B + A_L(s)(e + \eta), \quad (15)$$

$$0 \leq V^*(t, x) - V^a(t, x) \leq 2e^{Ls} B + A_L(s)(2e + \eta). \quad (16)$$

Consequently, exact evaluation, exact boundary data, and a zero feasible-action gap identify the optimal value and an optimal policy.

The theorem replaces an assertion about a global minimizer of a composite loss with a statement about the exact computational objects NBO estimates. It allows errors, does not require that a finite network class contain V^* , and bounds performance against all admissible policies covered by comparison. A small sampled

residual is not yet the uniform hypothesis of the theorem. Section 4.3 addresses that distinction.

4.1. A positive-weight recursive Bellman approximation

Let P_h^a be a nonnegative transition/interpolation operator on a finite state set, with row sums one after adjoining absorbing states. Boundary values are fixed. For a continuation array w , define $\Psi_h^a w$ pointwise as the scalar solution

$$z = P_h^a w + hf(x, a, z), \quad T_h w = \max_{a \in \mathcal{A}_h(x)} \Psi_h^a w. \quad (17)$$

For stationary calculations assume $f_y \leq -\lambda < 0$ on an invariant value domain, and require that all scalar roots remain in that domain. The map $z \mapsto z - hf(x, a, z)$ is strictly increasing with slope at least $1 + \lambda h$. Hence the root is unique and Ψ_h^a is order preserving and a sup-norm contraction with factor $q_h = (1 + \lambda h)^{-1}$. The maximum over feasible actions preserves these properties. A root bracket must contain the root; arbitrary clipping of utility values is not part of the definition.

For a finite horizon, the same implicit construction is used backward from the prescribed terminal array. Under the Lipschitz condition in Assumption 1, $hL < 1$ gives a one-step Lipschitz constant at most $(1 - hL)^{-1}$. Discount contraction is not assumed for a recursive parameter regime where it has not been verified. For a time-additive problem, the positive-weight backup $hr_0 + e^{-\rho h} P_h^a w$ is also consistent and monotone. It is the backup used by the NDU reference laboratory.

THEOREM 3—Discrete NBO certificate: *Suppose (17) is well defined on an invariant domain, with contraction factor $q_h < 1$. Let v_h^* and v_h^a denote the fixed points of T_h and T_h^a , respectively. For any neural value array v and deterministic feasible policy a , set*

$$\delta = \|v - T_h^a v\|_\infty, \quad \epsilon = \|T_h v - T_h^a v\|_\infty.$$

Then

$$\|v - v_h^a\|_\infty \leq \frac{\delta}{1 - q_h}, \quad \|v - v_h^*\|_\infty \leq \frac{\delta + \epsilon}{1 - q_h}, \quad \|v_h^* - v_h^a\|_\infty \leq \frac{2\delta + \epsilon}{1 - q_h}. \quad (18)$$

The certificate covers the stated finite state and action model. Continuous actions require an additional certified maximization or action-coverage error.

This result applies regardless of how the neural array was obtained. Neural projection need not itself be monotone. What matters is that the output satisfies the monotone Bellman equation to the displayed tolerance. The distinction makes it possible to use flexible representations while retaining a comparison-based acceptance test.

THEOREM 4—Viscosity-consistent neural Bellman approximation: *Consider stationary problems satisfying comparison in a bounded continuous class and the strict utility monotonicity used in Theorem 3. Suppose the positive-weight schemes are stable, locally consistent with the HJB operator, and attain the prescribed boundary data in the limit. The feasible action approximations become dense with vanishing maximization error. If neural arrays and feasible actors satisfy*

$$\delta_h + \epsilon_h = o(h), \quad (19)$$

then their interpolated values converge locally uniformly to the unique viscosity solution. Uniform convergence up to the boundary follows when boundary barriers give uniform boundary attainment. Their discrete policy-value arrays converge to the same limit. No assertion of convergence of the policy functions themselves is required.

Here stability, local consistency, comparison, and boundary attainment are numerical-analysis hypotheses, not consequences of calling a representation neural. The proof is given in the appendix. A finite-horizon version uses backward error accumulation: terminal error tends to zero and the sum of one-step errors tends to zero after multiplication by the one-step Lipschitz factors. This covers finite-horizon recursive problems without imposing stationary discount contraction.

For a diffusion with m Brownian factors, an interior semi-Lagrangian construction uses the $2m$ locations $x + hb \pm \sqrt{mh} \sigma e_j$, each with weight $1/(2m)$. Positive multilinear interpolation preserves monotonicity. For smooth test functions its interpolation contribution is $O(\Delta x^2/h)$; thus $\Delta x^2/h \rightarrow 0$ is a sufficient interior coupling condition. The mean and covariance of the cubature increments match those of the specified Brownian diffusion to first order. Stopping and reflection require consistent boundary transitions. Negative interpolation weights are not permitted in this construction.

4.2. Why an integrated differential residual does not select the value

Consider the exit problem $F(v') = 1 - |v'| = 0$ on $(-1, 1)$ with zero boundary payoff. Its value is $v^*(x) = |x| - 1$. The smooth functions

$$w_\epsilon(x) = \sqrt{1 + \epsilon^2} - \sqrt{x^2 + \epsilon^2}$$

satisfy the same boundary values and have vanishing integrated squared residual, but converge uniformly to $1 - |x|$. At zero the test function one touches the wrong limit from above and gives $F(0) = 1 > 0$, violating the viscosity subsolution inequality. The defect is not removed by exact Hamiltonian maximization.

On a grid with spacing h , the positive-weight exit backup is

$$(Tv)_i = -h + \max(v_{i-1}, v_{i+1}), \quad v_0 = v_N = 0. \quad (20)$$

It is an undiscounted shortest-exit problem, not an instance of the stationary contraction theorem. Backward propagation of the boundary values gives $v_i = |x_i| - 1$ exactly. At the peak of the wrong candidate, the *scaled* residual $|w_i - (Tw)_i|/h$ is close to two, rather than close to zero. The laboratory therefore distinguishes correct selection from small integrated differential error.

An explicit positive viscosity term is another possible regularization, provided comparison, boundary convergence, and approximation errors for the regularized equations are controlled before taking the viscosity to zero. Network width or spectral bias is not itself a positive viscosity parameter. Moreover, approximation of a C^2 function and its derivatives does not establish approximation of nonexistent classical derivatives at a kink.

4.3. Coverage and independent maximization

A verification distribution is declared independently of the actor's occupation measure. It includes economically relevant counterfactual states and boundary regions, not only states visited by a current policy. If a residual r has a known Lipschitz bound K on a compact domain and a validation set has fill distance d_{fill} ,

then

$$\|r\|_\infty \leq \max_{x \in \mathcal{V}} |r(x)| + Kd_{\text{fill}}. \quad (21)$$

The same bound can be applied to an action gap if its modulus is controlled. An empirical mean alone does not give this guarantee. If the sampling density is bounded below and the domain has a uniform interior-volume condition, a Lipschitz residual also admits an L^2 -to-sup estimate with exponent $1/(d+2)$, as derived in the supplement. Its dimension dependence is material.

An independent action solver must provide an *upper bound* on the unobserved maximization error. Trying a few alternative actions supplies only a lower bound on the true gap. Concavity with a duality gap, interval bounds, exhaustive enumeration of a finite action set, or an action net with a known Lipschitz modulus can provide the needed upper bound. An unverified local actor optimizer cannot certify itself.

4.4. Differential and discrete error accounts

For the stationary additive problem, write $f = r_0 - \rho v$, with $\rho > 0$, and let R_h restrict a candidate to the nodes of a positive-weight scheme. On interior nodes set

$$T_h^a w = q_h(P_h^a w + h r_0^a), \quad q_h = (1 + \rho h)^{-1}.$$

Stopping nodes instead use their prescribed payoff.

PROPOSITION 5—Transfer to a Bellman certificate: *Suppose the represented policy is feasible for the discrete action class and*

$$\sup_a \left\| h^{-1}(P_h^a R_h v - R_h v) - R_h \mathcal{L}^a v \right\|_{\infty, \text{int}} \leq \kappa_h.$$

If the differential evaluation residual is at most e and its feasible-action gap at most η , then, on the interior nodes,

$$\|R_h v - T_h^a R_h v\|_\infty \leq q_h h(e + \kappa_h), \quad \|T_h R_h v - T_h^a R_h v\|_\infty \leq q_h h(\eta + 2\kappa_h).$$

Boundary discrepancies and finite-action coverage errors are additional accounts.

The proof follows by adding and subtracting $R_h \mathcal{L}^a v$; the maximization inequality uses the consistency error twice. The converse is false for arbitrary oscillatory interpolants. For smooth interior functions, local cubature and interpolation estimates can bound κ_h ; they do not justify assigning zero error to a stopping or reflected boundary without checking its realization.

For finite-horizon maps with Lipschitz factors β_t , let δ_t bound evaluation error, ϵ_t bound improvement error, and B_N bound terminal error. Define

$$E_N = B_N, \quad O_N = B_N, \quad R_N = 2B_N, \quad \begin{cases} E_t = \delta_t + \beta_t E_{t+1}, \\ O_t = \delta_t + \epsilon_t + \beta_t O_{t+1}, \\ R_t = 2\delta_t + \epsilon_t + \beta_t R_{t+1}. \end{cases} \quad (22)$$

Backward comparison bounds policy-value error by E_t , optimal-value error by O_t , and policy regret by R_t . These arrays are computed, rather than replaced by a training loss, in the NDU experiment. Bounds at later time indices must use their own accounts, not necessarily the account at time zero. Representation and incomplete optimization enter δ_t ; action coverage enters ϵ_t ; the error of replacing the controlled diffusion by a scheme is separate. If a uniform continuous-to-discrete policy-payoff discrepancy is bounded by D_h , transferring a discrete regret bound adds $2D_h$. Without that estimate, the certificate remains a certificate for the stated discrete economy. In particular, this accounting does not establish the optimizer rate required by Theorem 4.

5. A TRAINED NEURAL BOUNDARY-VALUE PROBLEM

Consider a household with labor income, risky wealth, and a prescribed retirement payoff on exit. Before exit from $D = (0.5, 2.5)$,

$$dX_s = (0.12 + 0.02X_s - c_s) ds + 0.15X_s dW_s, \quad c_s \in [0.02, 0.6]. \quad (23)$$

Discount is $\rho = 0.08$, flow utility is $\log c$, and exit pays $g(x) = \log(0.12 + 0.08x)/0.08$. This is a specified stopped economy, not a claim about an unbounded-wealth problem. Income and the exit option destroy the homothetic reduction used in the portfolio calibration. Bounded coefficients, a compact positive consumption

interval, positive diffusion variance on the closed domain, and continuous endpoint data yield the usual stopped-domain comparison problem. A neural approximation must solve its nonlinear differential equation and both boundary conditions.

The critic has two hidden tanh layers of width sixteen and the lifting

$$v_{\xi}(x) = g(\ell) + \frac{g(u) - g(\ell)}{u - \ell}(x - \ell) + (x - \ell)(u - x)N_{\xi}\left(2\frac{x - \ell}{u - \ell} - 1\right).$$

The actor uses the same hidden architecture and a sigmoid into the consumption interval. Here the optimum is not forced to an endpoint by that representation. Independent verification nevertheless maximizes over the *closed* interval: for $p = v'(x)$,

$$c^*(p) = \begin{cases} \text{clip}(1/p, 0.02, 0.6), & p > 0, \\ 0.6, & p \leq 0. \end{cases} \quad (24)$$

Training imports the general differential core used for the earlier routing tests. It does not fit reference values, derivatives, or policy labels.

PROPOSITION 6—Continuous-domain economic certificate: *For a stopped discounted additive economy obeying comparison, let $v \in C^2(D) \cap C(\overline{D})$ and let a be admissible. Suppose throughout D that*

$$|\rho v - r_0^a - \mathcal{L}^a v| \leq e, \quad 0 \leq \sup_b (r_0^b + \mathcal{L}^b v) - (r_0^a + \mathcal{L}^a v) \leq \eta, \quad \|v - g\|_{\partial D} \leq B.$$

Then

$$\|v - V^a\|_{\infty} \leq B + \frac{e}{\rho}, \quad \|v - V^*\|_{\infty} \leq B + \frac{e + \eta}{\rho}, \quad 0 \leq V^* - V^a \leq 2B + \frac{2e + \eta}{\rho}. \quad (25)$$

Indeed, constant upper and lower shifts of v dominate the evaluation or optimality residual and the stopping data; comparison gives the first two inequalities and their sum gives the third. No reference-grid error is used in this argument.

The executable verifier covers the entire state interval by dyadic subintervals. Outward-rounded propagation through the stored network encloses v , v' , v'' , and the actor on every leaf. Equation (24) supplies a global action enclosure. We also

use the concavity bound

$$\log c^* - pc^* - (\log c - pc) \leq \frac{1}{2}C^2(1/c - p)^2, \quad C \geq \max\{c, c^*\}, \quad (26)$$

which follows by completing the square in the strong-concavity inequality on the segment joining the two actions. It does not assume an interior maximizer. Taking the better of this bound and a direct interval enclosure avoids dependency inflation. Terminal-free boundary errors are evaluated separately at both endpoints, including rounding in the lifting constants.

The certificate saves every leaf, verifies that their union is exactly $[0.5, 2.5]$, and binds the parameter and arithmetic files by hashes. Exponential and logarithmic enclosures use range reduction and polynomial remainder bounds rather than an assumed accuracy of a library transcendental. Its mathematical conclusion is conditional on the stated binary64 arithmetic contract; high-precision and derivative tests check the implementation but are not a formal machine proof. The construction does not estimate a global Lipschitz constant from samples.

5.1. *Controlled comparison and failures*

We use seeds 11, 29, and 47, each with 256 prespecified jittered collocation points. Eight outer rounds have a common cap of 800 L-BFGS inner iterations. NBO allocates each round sixty iterations to evaluation and forty to improvement; direct neural HJB uses the same critic and the global action rule in all one hundred iterations. A joint-loss ablation differentiates the sum of squared residual and negative Hamiltonian through both blocks. Line searches use different numbers of closures, so this is a matched iteration cap, not an assertion of equal floating-point work. Closure counts and elapsed times are saved.

An independent positive-rate upwind policy-iteration solver uses continuous-action maximization within each drift region. Reference grids have 320, 640, 1,280, and 2,560 intervals. Value and derivative comparisons use the finest grid; its refinement differences are not substituted for a rigorous discretization bound. The continuous certificate instead checks the neural functions directly. All three NBO and direct-HJB runs attain the declared interval-verification thresholds. The joint-

TABLE I
MULTILAYER CONSUMPTION: APPROXIMATION AND CONTINUOUS-DOMAIN VERIFICATION

Method	Seed	$\ v - V_h^*\ _\infty$	Sampled e	Uniform η	Regret bound
NBO	11	3.30×10^{-4}	6.18×10^{-4}	4.24×10^{-6}	0.0501
NBO	29	5.75×10^{-4}	0.0015	2.66×10^{-6}	0.0500
NBO	47	5.46×10^{-4}	0.0014	4.36×10^{-6}	0.0501
Direct HJB	11	4.14×10^{-4}	9.72×10^{-4}	0	0.0500
Direct HJB	29	3.55×10^{-4}	7.13×10^{-4}	0	0.0500
Direct HJB	47	6.00×10^{-4}	0.0017	0	0.0500
Joint loss	11	5.4142	0.5002	—	—
Joint loss	29	5.4435	0.5003	—	—
Joint loss	47	5.5677	0.5003	—	—

The comparison grid has 2,560 intervals on $[-.5, 2.5]$. The last two columns cover the entire continuous interval and consumption set $[.02, .6]$. All six certificates have uniform evaluation error at most .002 and endpoint error below 10^{-12} . Dashes indicate no accepted certificate, not zero error. Reference error and the sampled differential residual are not used as uniform bounds.

loss ablation does not: a residual near one half is accompanied by large value and action errors. Width, depth, and update-allocation variations are retained in the supplement, including unfavorable outcomes. The direct method is a strong comparator: these results do not establish that carrying an actor is uniformly faster or more accurate. They establish an end-to-end neural solve with an independently computable welfare bound, and identify a concrete failure of joint differentiation. Fitting and full-domain verification are separately timed because their computational burdens are different.

6. ANALYTICAL AND NONSMOOTH BENCHMARKS

The time-additive Merton model has wealth dynamics

$$dX = [(r + p(\mu - r))X - c]dt + p\sigma X dW, \quad U(c) = \frac{c^{1-\gamma}}{1-\gamma}.$$

For the stationary infinite-horizon benchmark, set $c = mX$, $V = KX^{1-\gamma}/(1-\gamma)$, and impose the usual admissibility and transversality requirements. Writing

$$p^* = \frac{\mu - r}{\gamma\sigma^2}, \quad R_* = r + p^*(\mu - r) - \frac{1}{2}\gamma(p^*)^2\sigma^2,$$

substitution and maximization give

$$m^* = \frac{\rho + (\gamma - 1)R_*}{\gamma}, \quad K = (m^*)^{-\gamma}. \quad (27)$$

With $\rho = .04$, $\gamma = 2$, $\mu = .08$, $r = .02$, and $\sigma = .2$, these numbers are $p^* = .75$ and $m^* = .04125$. These are stationary constants; an arbitrary finite-horizon problem with an unspecified bequest does not have the same constant consumption rule. A finite-horizon verification test can use the matching bequest $KX^{1-\gamma}/(1-\gamma)$, explicitly declared rather than silently omitted.

For $\mu = .01$ with no short sales, the unconstrained share is $-.125$ and the constrained optimum is zero at every positive wealth. Its effective return is r and $m^* = .03$. The maximized HJB remains an equality. The action KKT condition is an inequality at the boundary, but it does not turn this regular-control problem into an obstacle problem or create a wealth-dependent kink. The exit experiment in Section 4, rather than this homothetic corner solution, supplies the nonsmooth selection test.

6.1. Executed calibration and interpretation

The replication program trains a homothetic neural subfamily: a positive value coefficient, a sigmoid consumption-share output, and a projected portfolio-share output. These are neural output parameters in a known invariant representation, not an unrestricted multilayer approximation. The two Adam optimizers have learning rates .025 and .008; each outer iteration contains three critic updates and one actor update. There are 1,600 outer iterations and seeds 0, 1, 2. All attempted runs, including their coefficient and policy traces, are saved. These calibrations optimize the stationary coefficient equation and then check the resulting coefficient against the matching bequest; the coefficient error is a boundary error in that

TABLE II

CORRECTED PORTFOLIO CALIBRATION: ANALYTICAL TARGETS AND EXECUTED BLOCK UPDATES

Model	Target m	Target p	Runs	Maximum policy error
Time-additive Merton	.04125	.75000	3	5.6×10^{-16}
No-short-sale corner	.03000	.00000	3	2.3×10^{-16}
Recursive matching-bequest test	.04550	.30000	3	7.0×10^{-17}

Policy error is the maximum absolute error in the two constant policy outputs, maximized over all three seeds. Calculations use CPU float64 in the declared homothetic subfamily. Machine-level calibration accuracy is not evidence about an unrestricted network, a GPU runtime, or a general recursive infinite-horizon problem.

finite-horizon interpretation, not a boundary condition hard-imposed during these particular updates.

Table II reports quantities generated by the new code. The normalized policy-evaluation residual is below 4×10^{-16} in these runs. These results establish that the revised graph and formulas agree in the exactly representable laboratory. They do not establish a broad advantage over a direct analytical calculation. In the exit laboratory, grids with 40, 80, 160, and 320 intervals recover the grid value to within 3×10^{-15} ; the wrong smooth sequence has scaled residual between 1.984 and 1.999. All grid nodes, not a selected interior sample, enter that diagnostic.

7. RECURSIVE UTILITY

Let $a = 1 - 1/\psi$, $b = 1 - \gamma$, and $Y = bV > 0$, with $\psi \neq 1$ and $\gamma \neq 1$. We use the difference-form aggregator

$$f(c, V) = \frac{\rho}{a} \left[c^a (bV)^{1-a/b} - bV \right]. \quad (28)$$

Its consumption derivative is

$$f_c = \rho c^{a-1} (bV)^{1-a/b} > 0. \quad (29)$$

When $\gamma = 1/\psi$, one has $a = b$ and $f = \rho(c^b/b - V)$, a positively normalized time-additive CRRA aggregator. The factor b must not multiply the consumption term outside its power. Doing so reverses consumption monotonicity when $\gamma > 1$. This

distinction follows the difference-form convention discussed by [Herdegen et al. \(2021\)](#).

For a homothetic candidate $V = KX^b/b$, the maximization equation becomes

$$0 = \sup_{m,p} \left\{ \frac{\rho}{a} [m^a K^{-a/b} - 1] + r + p(\mu - r) - m - \frac{1}{2} \gamma p^2 \sigma^2 \right\}. \quad (30)$$

The consumption first-order condition is $\rho m^{a-1} K^{-a/b} = 1$. Substitution yields

$$p^* = \frac{\mu - r}{\gamma \sigma^2}, \quad m^* = \psi \rho + (1 - \psi) \left[r + \frac{(\mu - r)^2}{2\gamma \sigma^2} \right], \quad K = [\rho (m^*)^{a-1}]^{b/a}. \quad (31)$$

For the retained parameters $\gamma = 5$, $\psi = 1.5$, $\rho = .04$, $\mu = .08$, $r = .02$, $\sigma = .2$, this gives $(m^*, p^*) = (.0455, .3)$.

7.1. A well-defined validation problem at the retained parameters

The numerical calibration uses (31) as a finite-horizon matching-bequest test, not as an unqualified infinite-horizon theorem. Choose $T > 0$, stop wealth on leaving $[\underline{x}, \bar{x}]$ with $0 < \underline{x} < \bar{x} < \infty$, and impose the candidate Kx^b/b on the stopping boundary and at T . Consumption shares are in a compact positive interval containing .0455, and portfolio shares are in a compact interval containing .3.

For these retained parameters, $a > 0$ and $b < 0$. Since c ranges over a compact positive interval, there exist $0 < \underline{Y} < \bar{Y} < \infty$ such that $f(c, \underline{Y}/b) \leq 0$ and $f(c, \bar{Y}/b) \geq 0$ for every feasible c . Indeed, $c^a Y^{-a/b}$ tends to zero as $Y \downarrow 0$ and to infinity as $Y \uparrow \infty$. Enlarge the interval so that it contains the boundary candidate. The constants $\bar{Y}/b$ and $\underline{Y}/b$ are respectively lower and upper barriers for the backward utility equation. Scalar BSDE comparison therefore keeps every policy utility in this interval. The aggregator is Lipschitz there. The smooth homothetic candidate maximizes the Hamiltonian and matches all stopping payoffs; Theorem 2 identifies it as the value of this specified finite-horizon problem.

The choice of a matching bequest is an analytical validation device. It does not establish that the same candidate is selected by every infinite-horizon recursive formulation. A separate infinite-horizon application must check its utility-process class and transversality, particularly when risk aversion and intertemporal-

complementarity parameters straddle one (Herdegen et al., 2021). The paper retains that economic application, but does not use a scalar root to bypass its selection issue.

A critic can enforce $bV > 0$ through $bv_{\xi} = \exp(N_{\xi})$. With a positive terminal target, logarithmic interpolation between bG and the neural output enforces both the terminal condition and the power domain. Additional barriers or bounded transformations enforce a declared invariant interval. Such transformations are stated in the architecture, not imposed retrospectively after a failed evaluation.

7.2. Stability and baseline design

The recursive experiment in Table II tests the corrected sign, coefficient, and block graph. It does not compare independent DDPG training trajectories. The reviewed manuscript's constructed oscillating curve and manually entered coordinates remain in the archival source, but have no evidentiary role in this revision. A matched neural comparison must use the same utility process, domain, bequest, action restrictions, training budget, and held-out evaluation quantities. It must report all failed seeds and value-domain violations. Neither a zero-centered target nor three successful homothetic calibrations establishes general relative stability.

8. ENDOGENOUS PREFERENCES AND PORTFOLIO CHOICE

The endogenous-preference application retains its internal adjustment margin. Let u be risk aversion, X wealth, and θ adjustment effort. On a declared interval $[\underline{u}, \bar{u}] \subset (1, \infty)$, specify the reflected process

$$du = \theta dt + \sigma_u dW^u + dL^u - dL^{\bar{u}}, \quad (32)$$

$$dX = [(r + p(\mu - r))X - c]dt + p\sigma_X X dW^X, \quad d\langle W^u, W^X \rangle_t = \varrho dt. \quad (33)$$

The nondecreasing local-time regulators act only at their respective boundaries. This supplies the state mechanism that a bounded adjustment actor does not supply. The numerical reference uses $[1.2, 3]$, so the utility singularity at $u = 1$ is never reached. In an alternative positive-state model one could specify a multiplicative

or transformed diffusion, but that would be a different primitive and must be identified as such.

8.1. Cardinal utility and the objective

Fix a reference consumption unit $\bar{c} > 0$. The baseline flow utility is

$$U(c, u) = \frac{(c/\bar{c})^{1-u}}{1-u}, \quad r_0(c, u, \theta) = U(c, u) - \frac{1}{2}k\theta^2. \quad (34)$$

The reference calculation takes $\bar{c} = 1$, preserving the unnormalized CRRA family of the original application. The agent maximizes

$$\mathbb{E}_{t,u,X} \left[\int_t^\tau e^{-\rho(s-t)} r_0(c_s, u_s, \theta_s) ds + e^{-\rho(\tau-t)} G(u_\tau, X_\tau) \right]. \quad (35)$$

Both the payoff and the adjustment cost have the same time-relative discount convention. The terminal/exit payoff and the state domain are stated for each calculation.

A useful family for normalization sensitivity is

$$U_\kappa(c, u) = \frac{(c/\bar{c})^{1-u} - 1}{1-u} + \kappa(u).$$

The baseline corresponds to $\kappa(u) = 1/(1-u)$. The log-continuous normalization corresponds to $\kappa = 0$ and is an economically different preference-formation model. A calibration of κ fixes the cardinal value of a reference consumption stream at each preference state. The choice is not identified by risk-aversion curvature alone.

In particular,

$$U_u(c, u) = \frac{(c/\bar{c})^{1-u} \{1 - (1-u) \log(c/\bar{c})\}}{(1-u)^2}. \quad (36)$$

For $c/\bar{c} > 1$ and $u > 1$ this derivative is positive. At $c = 2$, $u = 2$, $\bar{c} = 1$, it is approximately .84657. This is a derivative of flow utility, not a sign theorem for the endogenous value derivative V_u . The agent's full incentives also depend on future consumption, boundary behavior, and the adjustment technology.

8.2. *Hamiltonian, constrained policies, and economic implications*

The HJB equation is

$$\begin{aligned} -V_t = \sup_{c,p,\theta} \{ & U(c,u) - \frac{1}{2}k\theta^2 - \rho V + \theta V_u + [(r + p(\mu - r))X - c]V_X \\ & + \frac{1}{2}\sigma_u^2 V_{uu} + \frac{1}{2}p^2\sigma_X^2 X^2 V_{XX} + \varrho\sigma_u\sigma_X p X V_{uX} \}. \end{aligned} \quad (37)$$

Reflection gives $V_u = 0$ at the two preference boundaries, in the appropriate boundary-value sense. The correlation term is retained with its correct coefficient.

For an interior optimum with $V_X > 0$ and $V_{XX} < 0$, the first-order conditions are

$$c^* = \bar{c} [\bar{c} V_X]^{-1/u}, \quad (38)$$

$$\theta^* = V_u / k, \quad (39)$$

$$p^* = -\frac{V_X(\mu - r)}{X V_{XX} \sigma_X^2} - \frac{\varrho \sigma_u V_{uX}}{X V_{XX} \sigma_X}. \quad (40)$$

For independent box constraints, strict concavity in the corresponding action permits projection of these unconstrained maximizers onto their intervals. If V_{XX} has the wrong sign or constraints couple the actions, a joint feasible maximization is required instead. The formula for p must not be used where its denominator vanishes.

Equation (39) gives a conditional shadow-price interpretation of internal adjustment. Holding V_u fixed, a larger k reduces its magnitude. Across economies, however, V_u itself depends on k , so this is not a global policy comparative static. A valid unconditional implication is that the optimized value is weakly decreasing in k : every fixed admissible policy loses $\frac{1}{2}(k_2 - k_1)\mathbb{E} \int e^{-\rho s} \theta_s^2 ds$ when the cost rises. When an envelope derivative exists,

$$\partial_k V^k = -\frac{1}{2}\mathbb{E}^{a^k} \int_t^\tau e^{-\rho(s-t)} \theta_s^2 ds \leq 0. \quad (41)$$

The sign of the hedging component in (40) remains conditional on V_{uX} and the curvature of wealth value. A negative correlation alone does not determine it.

8.3. Independent two-state reference and identification

The executable reference solves the finite-horizon problem with $T = 1$, $u \in [1.2, 3]$, wealth stopped at $[.5, 2]$, and $G(u, X) = X^{1-u}/(1-u)$ at exit and terminal time. Parameters are $(r, \mu, \sigma_X, \sigma_u, \rho) = (.02, .08, .2, .08, -.3, .04)$. The finite action set is the product of $c/X \in \{.02, .05, .1\}$, $p \in \{0, .375, .75\}$, and $\theta \in \{-.15, 0, .15\}$. A four-point Brownian cubature with positive interpolation weights performs backward induction. The wealth boundary is imposed at every time step; preference overshoots are reflected. This reference defines a finite-action numerical approximation, not continuous-action optimality.

At $(u, X) = (2.1, 1)$, the original 17×25 state grid and 20 time steps give values -11.915211 and -11.956983 for $k = 1$ and $k = 5$. The 25×37 grid and 40 time steps give -11.866768 and -11.908738 . Both select $(c/X, p, \theta) = (.1, .75, -.15)$ at that point. The approximately .048 refinement change is reported separately from training error. The higher-cost value is weakly lower at every recorded node and time. These original references remain useful consistency checks; Section 8.4 supplies actual neural fits and shows how strongly the finite action domain affects their economic comparison.

The model also specifies what a structural estimation exercise would require. For observations $Y_j = \mathcal{M}(S_{t_j}, a_{t_j}; \vartheta) + \varepsilon_j$, let $m(\vartheta)$ denote selected population moments of the induced observation process. Local identification requires a rank condition on $D_{\vartheta}m$, after accounting for latent preference states, normalization parameters, and observation noise. Variation of a simulated policy with k alone is insufficient. Numerical value and policy errors must additionally be propagated into moment errors. The endogenous-preference application is therefore retained as an economic model with an explicit internal margin, not used as evidence that its structural parameters have already been identified from data.

8.4. Neural preference policies and the role of the action oracle

We compute the original preference-adjustment economy with the cardinal utility, shock correlation, reflected preference state, and wealth stopping payoff specified above. The main grid has 17 preference points, 25 log-wealth points, and 20 time

steps over a unit horizon. Adjustment cost is $k = 1$. The expanded control box is

$$m = c/X \in [.02, .30], \quad p \in [0, 1.5], \quad \theta \in [-.45, .45].$$

The inherited finite-action comparison uses the 175 triples formed by five consumption shares, five portfolio shares, and seven adjustment efforts, including all endpoints. This is the same constrained economy across the matched finite methods, not a change of utility normalization.

Two experiments distinguish action representation from action optimization. First, the R7 finite study trains two-hidden-layer, width-32 tanh critics and categorical actors. At each backward step it computes the complete 175-action payoff vector using the current continuation critic. Its exact argmax provides a label, and the vector also supplies the actor’s expected action-loss objective. Thus *the training oracle is exhaustive*, even when the final actor is accurate. It is a useful neural policy-representation experiment, but not evidence that the actor avoided Bellman maximization.

Second, R8 trains a continuous-output actor directly through its own positive-weight payoff. It never receives an all-action value vector or an exact argmax label. The actor receives 160 Adam updates per time level; the continuation critic receives 320 Adam updates and up to 300 L-BFGS iterations, with failed polishing steps rolled back. The actor-free direct comparator optimizes eight independent bounded controls at every state for the same 160 iterations, followed by the same critic fit. Every restart and rejected action query is charged. Neither procedure invokes a Riccati solution, an analytically invariant policy class, or the independently computed reference during training.

Finite-action neural policies. The raw R7 actor is the primary neural representation result. Its full dynamic policy loss is below the declared .1 target for each fixed seed. A top-four shortlist and a separately reported correction map further reduce the loss. The correction map is not part of the raw network. The complete finite policy, not only one-step deviations against a fitted critic, is evaluated by independent backward recursion. These results repair the earlier R6 actor failures, which remain in the historical numerical record. They do not reverse the cost comparison: the

TABLE III
FINITE PREFERENCE POLICIES: R7 EVIDENCE

Method	Seed	Raw loss	Final loss	Value error	Seconds	Guard (%)
NBO	11	0.0445	0.0261	0.0924	30.59	0.294
Direct	11	0.0181	0.0181	0.0790	24.79	0.000
NBO	29	0.0500	0.0260	0.0814	30.49	0.332
Direct	29	0.0263	0.0263	0.0551	25.19	0.000
NBO	47	0.0500	0.0263	0.1325	29.89	0.230
Direct	47	0.0264	0.0264	0.0824	25.17	0.000

Loss bounds cover every stored state and time of the same 175-action economy. Seconds include training and the three reported policy verifications; classical backward induction averages 0.102 seconds. A raw actor, shortlist, and corrected policy are distinct objects. Utility losses are absolute units.

matched direct finite-Bellman fit is faster and more accurate on average, and classical backward induction is substantially cheaper on this small grid. Tables report those comparisons without selecting a favorable seed. The actor’s approximation cost is part of the result, not an argument for discarding the economic application.

Continuous-action optimization without exact labels. The initial continuous actor and direct-gradient pilot fail the .1 deviation diagnostic. The actor can saturate at a portfolio bound while a lower portfolio share improves continuation value. Four actor heads with output reinitialization do not cure that failure in the retained pilot. We therefore execute a disclosed local-improvement variant: compare the actor proposal with eight box corners, then perform two coordinate-search sweeps at each of eight decreasing step sizes. The same local search is applied to the direct comparator. An actor-free ablation omits all gradient action updates and starts the identical search from feasible proposals. This ablation measures how much of the hybrid result comes from the search itself. The table distinguishes three quantities. The raw-network loss against the finite reference tests the actor alone. The augmented finite-deviation gain permits all 175 inherited actions and the learned continuous action at every state and time; it is a *lower bound* on missed continuous-action gains. The complete action-box upper bound in Proposition 1

TABLE IV
CONTINUOUS-CONTROL TRAINING WITHOUT EXACT MAXIMIZING LABELS

Method	Seed	Raw diagnostic	Hybrid gain	Train (s)	Queries (10^6)
Actor + search	11	0.2596	0.0617	15.01	2.261
Actor + search	29	0.0733	0.0188	28.49	2.261
Actor + search	47	0.2597	0.0475	26.46	2.261
Direct + search	11	0.1554	0.0303	32.02	11.841
Direct + search	29	0.1566	0.0203	32.80	11.841
Direct + search	47	0.1554	0.0251	30.52	11.841
Search only	11	5.5370	0.0617	13.97	0.961
Search only	29	7.2093	0.0616	13.72	0.961
Search only	47	8.0316	0.0616	12.87	0.961

Raw diagnostic is the largest finite-reference payoff minus the raw policy payoff. Hybrid gain is the augmented finite-deviation gain, a lower bound on continuous-action regret, not a certificate. All methods use zero exhaustive training backups and zero exact argmax training labels. Search-only omits action-gradient updates. Training, reference, and full-cover verification costs are separate.

is a separate verification result. A small augmented gain is not inserted into that proposition as if it were a uniform upper bound. Local changes occur at nearly every active state-time node in the principal hybrid runs, so this experiment does not describe a sparse correction map or an actor-only controller. The action cover includes binding constraints and wealth exits. Its unresolved boxes remain in the bound when the declared budget is exhausted. The resulting upper array can audit the actor, the direct optimizer, and the local-search ablation without retraining. This produces an operational error account on a multidimensional state grid with a continuous, nonconcave action objective. It does not establish that all runs meet the tighter .1 complete-action target, nor that the actor dominates the direct method at matched economic accuracy.

Alternative representations and economic units. Additional same-model comparisons use a gated neural continuation architecture and tensor Chebyshev projection. The gated method uses DGM-style state-dependent gates (Sirignano and Spiliopoulos, 2018), but here it is fitted to finite Bellman targets, not called a mesh-free

TABLE V

FULL CONTINUOUS-ACTION POLICY BOUNDS ON THE TWO-STATE NODAL ECONOMY

Policy	Seed	Uniform bound	Center bound	Eval. (s)	Bound $\leq .1$
Actor + search	11	0.1294	0.0601	0.22	No
Actor + search	29	0.1227	0.0583	0.22	No
Actor + search	47	0.1301	0.0608	0.25	No
Direct + search	11	0.1153	0.0604	0.23	No
Direct + search	29	0.1156	0.0586	0.22	No
Direct + search	47	0.1203	0.0609	0.25	No
Search only	11	0.1196	0.0647	0.24	No
Search only	29	0.1179	0.0586	0.23	No
Search only	47	0.1209	0.0610	0.22	No

The full action box is covered at all 425 states and 20 times. The common optimal upper envelope costs 1624.59 seconds across the retained implementations and 40.96 million action-box evaluations. Each row adds its own interval policy evaluation. Unresolved leaves retain their upper bounds. No continuous-state/time error is set to zero.

differential DGM solver. The projection exploits the same two-state geometry and imposes the same wealth-boundary lifting. Degrees, parameter counts, all-action costs, failures, and actual accuracy appear in the supplement. These are executed post-review comparisons, not results attributed to the earlier unexecuted development protocol. A generic semilinear BSDE comparator is not substituted for this controlled-covariance, stopped problem without reformulating its second-order control term.

All losses are in the specified cardinal utility units. The optimal finite-model value at the center is approximately -4.8410 , and the initial value range is approximately $[-9.1599, -.1250]$. A loss of $.1$ is about 2.066 percent of the center's *absolute utility magnitude*; this scaling is not a consumption-equivalent welfare percentage. Endogenous curvature and adjustment costs make a common consumption-equivalent conversion a separate economic calculation.

Action bounds and controlled refinements. The inherited sensitivity study retains both adjustment-cost values, the original 27-action truncation, 175 and 1,053 ac-

tions in the expanded box, a wider action box, wider wealth bounds, and finer space–time grids. Consumption and portfolio upper bounds remain frequently active. Expanding the box changes the constrained economy; refining actions inside one box does not. R8 additionally evaluates the *same frozen continuous policy*, bilinearly interpolated in state and held on its original time schedule, on three nested state–time levels. This isolates changes in policy evaluation from retraining. The supplemental differences are observed refinement diagnostics, not an estimated order promoted to a continuous-time error bound. The original continuous control problem, its utility normalization, and its comparative-static questions remain the economic objects of interest.

9. SOPHISTICATED TEMPORAL SELVES

To define present bias in continuous time, fix the discount kernel

$$h(s) = \beta e^{-\rho s} + (1 - \beta) e^{-(\rho + \kappa)s}, \quad 0 < \beta \leq 1, \quad \kappa > 0. \quad (42)$$

It satisfies $h(0) = 1$ and discounts the immediate future more heavily than the distant future when $\beta < 1$. It is a finite-duration continuous-time specification; it is not the assertion that multiplying current utility by β creates present bias.

For a candidate Markov policy a^* , define the two continuation functions

$$W_j(t, x) = \mathbb{E}_{t,x}^{a^*} \left[\int_t^T e^{-\rho_j(s-t)} u(a_s^*) ds + e^{-\rho_j(T-t)} G(S_T) \right], \quad (\rho_1, \rho_2) = (\rho, \rho + \kappa). \quad (43)$$

Write $V = \beta W_1 + (1 - \beta) W_2$. Each component evaluates the *same* future policy:

$$0 = (W_j)_t + \mathcal{L}^{a^*} W_j + u(a^*) - \rho_j W_j, \quad W_j(T) = G. \quad (44)$$

The current self’s spike-deviation condition is

$$a^*(t, x) \in \arg \max_{a \in \mathcal{A}(t, x)} \{u(a) + \mathcal{L}^a V(t, x)\}. \quad (45)$$

The deviations change both current flow utility and the transition of the continuation state. They do not hold all dynamic effects at the equilibrium action. The

appendix derives (45) from a deviation on $[t, t + \epsilon]$ followed by a^* , thereby fixing the equilibrium concept.

A temporal-self NBO uses two policy-evaluation critics for (44) and one actor for (45). The current actor does not optimize the sum of future selves' parameter objectives. A target actor can stabilize a numerical iteration, but its agreement with the current actor is not an equilibrium certificate. The relevant diagnostics are component evaluation errors and one-shot deviation gains. The benchmark $\beta = 1$ collapses to the usual exponentially discounted problem.

9.1. An equilibrium calculation with endogenous continuation values

For deterministic wealth $dX = (rX - c)dt$, CRRA utility with $\gamma > 1$, and an infinite horizon, consider $c = mX$. Set $b = 1 - \gamma$ and $\lambda_j(m) = \rho_j - b(r - m) > 0$. Then

$$W_j(X) = \frac{m^b}{\lambda_j(m)} \frac{X^b}{b}, \quad 1 = m \left[\frac{\beta}{\lambda_1(m)} + \frac{1 - \beta}{\lambda_2(m)} \right]. \quad (46)$$

The expression on the right is strictly increasing in m on its admissible interval, starts at zero, and diverges at the first pole for $\beta > 0$. The equilibrium root is unique in this homothetic class. Its derivative with respect to β is negative because $1/\lambda_1 > 1/\lambda_2$. Thus stronger present bias raises consumption in this specified equilibrium, not merely at a fixed shadow price.

With $r = .02$, $\rho = .04$, $\gamma = 2$, and $\kappa = .2$, the roots for $\beta = 1, .9, .7, .5$ are respectively .030000, .031372, .034606, and .038760. The code independently optimizes finite-duration deviations and records the gain divided by duration as the duration shrinks. For $\beta = .7$, this quantity declines from approximately 4.60×10^{-5} at duration .1 to 4.66×10^{-9} at duration .001. This is a diagnostic of the derived spike equilibrium. It does not assert that every finite-duration commitment deviation is unprofitable, which would be a different solution concept.

10. DYNAMIC GAMES

The dynamic Cournot application retains firm-specific capital and investment.

Firm i has

$$dK^i = (I^i - \delta K^i)dt + \sigma_K K^i dW^i, \quad q_i = AK^i,$$

and payoff rate

$$r_i = P_N(K)q_i - I^i - \frac{1}{2}b_i(I^i)^2, \quad P_N(K) = \max \left\{ 0, P_0 - \frac{1}{M_N} \sum_j q_j \right\}.$$

For each experiment, the market-size normalization M_N , capital domain, investment constraints, covariance, terminal/liquidation values, and horizon are part of the primitives. Keeping $M_N = 1$ is a fixed-market comparison. Taking $M_N = N$ is a growing-market comparison. They do not have the same competitive-limit interpretation.

Given rivals' Markov policies a^{-i} , a player's best-response equation maximizes only its own action. In the stationary additive case,

$$\rho_i V_i = \sup_{a_i \in \mathcal{A}_i} \{r_i(x, a_i, a^{-i}(x)) + \mathcal{L}^{a_i, a^{-i}} V_i\}. \quad (47)$$

A Markov-perfect equilibrium is a feasible profile satisfying these simultaneous unilateral conditions, with the same boundary and admissibility conventions. A zero policy-evaluation residual can hold for any consistently valued profile and therefore does not identify this equilibrium.

10.1. Player-specific differentiation and deviation bounds

Player i 's critic is trained with every policy frozen. Player i 's actor then differentiates only

$$-\mathbb{E}_V \mathcal{H}_i^{a_i(\phi_i), \text{sg}(a^{-i})} [\text{sg}(V_i), \text{sg}(DV_i), \text{sg}(D^2 V_i)]. \quad (48)$$

The updated parameter is ϕ_i , not the full vector of all players' parameters. Rival actions are detached, and player j 's loss never contributes to player i 's actor gradient. An explicit regression test exercises this graph.

For each player, define the unilateral Hamiltonian gap by maximizing $\mathcal{H}_i$ over a_i while keeping rivals fixed. Applying Theorem 2 to that best-response problem bounds its finite-horizon exploitability by $2e^{L_i T} B_i + A_{L_i}(T)(2e_i + \eta_i)$. In a discounted finite-state game, Theorem 3 gives $(2\delta_i + \epsilon_i)/(1 - q_i)$. These are player-specific bounds; a sum of critic losses is not a substitute. They remain valid without uniqueness or convergence of the simultaneous policy-update dynamics, provided the reported profile satisfies the stated errors and best-response comparison hypotheses.

The static Cournot diagnostic makes the distinction visible. With payoffs $q_i(1 - q_i - q_j)$, symmetric Nash output is $1/3$ per firm. Joint-profit maximization gives a symmetric allocation $1/4$. At the latter profile, a firm optimally deviates to $3/8$, earning a gain of $1/64$. A joint objective can therefore be perfectly optimized and still fail a unilateral equilibrium test. This arithmetic is included as a mandatory regression check.

10.2. Symmetry and market scaling

Identical initial weights do not prove equilibrium uniqueness or permutation equivariance. An equivariant actor can be specified using own capital and a permutation-invariant aggregation of rival states. Even then, one must check unilateral deviations rather than infer equilibrium from architectural symmetry. Multiple starting profiles and unsuccessful best-response searches belong in a numerical report.

Under fixed market size, positive constant capital per firm cannot be called the competitive limit while total output tends to infinity and the truncated price tends to zero. Maintaining that capital entails investment and adjustment costs. Under $M_N = N$, positive capital per firm may instead be compatible with a growing market, but it describes a different sequence of economies. The original unsupported capital table remains in the archive. Section 10.3 replaces its evidentiary role with a fully specified finite dynamic game and independent best-response calculations, without asserting convergence of unrestricted neural game training.

10.3. *Neural dynamic capital policies and full unilateral deviations*

For two firms, we retain the capital game with $A = 1$, $P_0 = 2$, $\delta = .1$, $\sigma_K = .15$, $\rho = .04$, $T = 3$, investment in $[0, .8]$, and capital in $[\cdot 05, 1.5]^2$. Terminal or joint-domain exit pays $.5K_i$. Baseline adjustment coefficients are $b_1 = b_2 = 1$; the earlier asymmetric experiment has $b_2 = 1.2$. The principal neural grid has 17×17 states, 30 time steps, and 11 actions per player. Positive cubature and interpolation define the finite game. Market normalizations $M_2 = 1$ and $M_2 = 2$ are distinct economies, not evidence for an uncomputed large-population limit.

At each time and state, the R7 calculation trains separate critics and actors. It constructs both full 11×11 continuation-payoff matrices. The training labels are a pure one-step equilibrium when one exists and a minimum-maximum-defect pair otherwise. The actor's own objective is differentiated with the rival action and continuation critic fixed. Thus this study actually trains neural strategic policies, but it also pays for an exhaustive joint-action training oracle. Raw policies, three-by-three shortlist policies, and any corrected policies are distinct stored arrays.

After freezing the entire profile, an independent backward optimization permits every own action at every state and time against the fixed rival policy. Its value minus the profile payoff is the unilateral exploitability map. This is not a one-period test with the approximate continuation held fixed. The raw and final profiles meet the .1 absolute-profit target in every principal R7 case; the full-oracle guard is unused in those cases. Five state-time nodes across the six neural runs lack a pure one-step equilibrium for the fitted continuation matrices. Their locations and defects are reported in the supplement. A minimum-defect selection at those nodes is not called an exact equilibrium. Nevertheless, the complete dynamic best-response bound remains valid for the resulting profile: its proof does not assume that every fitted one-step game has a pure solution. Moreover, for fixed Markov rivals, randomized or history-dependent own deviations cannot improve on the finite Markov best-response value, because each conditional action mixture is bounded by its largest pure-action payoff. The calculation therefore checks such unilateral deviations as well; it does not compute a mixed-strategy equilibrium profile or a continuous-action game.

TABLE VI

R7 NEURAL DYNAMIC-GAME PROFILES AND FULL BEST RESPONSES

Market	Seed	Raw max	Firm 1	Firm 2	Missing	Seconds
1	11	0.0259	0.0164	0.0252	3	52.43
2	11	0.0241	0.0255	0.0163	1	56.13
1	29	0.0300	0.0288	0.0296	0	52.44
2	29	0.0626	0.0624	0.0547	1	57.23
1	47	0.0190	0.0081	0.0183	0	52.53
2	47	0.0101	0.0074	0.0113	0	56.98

All bounds cover every stored finite state, time, and unilateral action, including deviations at the reported missing pure stages. The full-oracle guard is unused in these principal runs. Profit losses are absolute units; market normalizations are different economies.

R8 adds actor-free neural continuation fitting and classical backward selection for both market normalizations. The neural comparator allocates 480 critic Adam updates per time level rather than R7's 320 critic plus 160 actor updates, with the same L-BFGS cap. Counts and timings reveal the difference between nominal update matching and matched wall time. Both methods receive their own independent full dynamic best responses. The earlier six classical game cases, asymmetric costs, alternative tie rules, boundary frequencies, coarse counterexample, and finer-grid results remain in the supplement and raw record. The finite-game result is a numerically evaluated approximate equilibrium with a specified profit-unit bound. It is not inferred from stationarity of a summed actor loss or the static 1/64 diagnostic. State interpolation, continuous investment, diffusion timing, and continuously monitored exits remain separate terms in a continuous-game error account.

11. SCALABILITY AND THE DIFFUSION TRACE

The original multi-stock consumption-savings specification has additive utility, independent transitions, and no common resource constraint. It decomposes as $V(k) = \sum_j v_j(k_j)$. A separability-aware comparator solves the one-dimensional components, and identical components can be reused. An unreduced sparse grid is not an appropriate sole comparator for that economy.

TABLE VII

R8 ACTOR-FREE DYNAMIC-GAME COMPARISONS

Method	Market	Seed	Firm 1	Firm 2	Train (s)	Verify (s)
Classical	1	11	0.0000	0.0000	0.22	6.40
Classical	2	11	0.0000	0.0000	0.23	6.29
Direct	1	11	0.0223	0.0273	37.43	6.09
Direct	1	29	0.0199	0.0282	37.68	6.18
Direct	1	47	0.0223	0.0409	36.03	6.26
Direct	2	11	0.0091	0.0694	40.43	5.75
Direct	2	29	0.0306	0.0185	43.07	6.34
Direct	2	47	0.0206	0.0358	40.47	6.08

The direct method fits two continuation networks with the same class and 480 critic Adam updates, versus R7's 320 critic and 160 actor updates. Classical selection uses no neural approximation. Every method receives a new full dynamic best response. R7 and R8 timings were measured in their recorded environments; cross-environment ratios are not a hardware-controlled speed experiment.

To test a coupled generator, the new laboratory instead uses

$$dX = (AX + a)dt + \Sigma dW, \quad \sup_a \mathbb{E} \int_0^\infty e^{-\rho t} [-X^\top QX - a^\top a] dt. \quad (49)$$

The drift is a nonnormal dense matrix $A = -.7I + .06Z/\sqrt{d}$, where Z is generated by the fixed model seed $991 + d$. The payoff matrix is $Q = \text{diag}(1, \dots, 2) + .311^\top/d$. The code verifies that $AQ \neq QA$, so the common independent-coordinate reduction of the original benchmark is unavailable. All methods still exploit the declared quadratic structure. A Riccati solver is therefore the appropriate strong reference, rather than a deliberately unreduced general grid.

For $V = -x^\top Px - c_0$ and $a = -Kx$, policy evaluation solves

$$(A - K - \rho I/2)^\top P + P(A - K - \rho I/2) = -(Q + K^\top K), \quad c_0 = \text{tr}(\Sigma \Sigma^\top P)/\rho. \quad (50)$$

Hamiltonian improvement gives $K = P$. A positive semidefinite P is representable by a signed sum of squared linear neural units; the actor is linear. The exact block laboratory uses a Lyapunov solve for this critic and exact actor improvement. It is

an exactly representable NBO subfamily and a matrix-identity test, not evidence that generic stochastic neural optimization has the same convergence rate.

Across dimensions 2, 5, 10, 20, 50 and three stabilizing actor initializations, the block iteration takes six steps. The maximum error in P , in operator norm, is below 10^{-13} relative to an independent algebraic Riccati solver. The policy-evaluation coefficient residual is below 10^{-12} . On the unit Euclidean ball, the actor gap is bounded by $\|K - P\|_2^2$, and the residual bound follows from the operator norm of the quadratic coefficient error. The program saves matrices, iteration traces, and CPU wall times for both methods. These very short, single-process timings are diagnostics and do not establish a fixed-accuracy runtime ranking.

11.1. *Trained, coupled nonquadratic capital economies*

To separate neural scaling from exact quadratic algebra, consider d positive capital stocks and write $Y_i = \log X_i$. Consumption is $c_i = m_i X_i$, with $m_i \in [.02, 2]$, and

$$dY_i = \left[.10 + .15 \tanh((B_d Y)_i) - m_i - \frac{1}{2}(.15^2 + .10^2) \right] dt + .15 dW_i + .10 dW_0, \quad (51)$$

$$r_0(Y, m) = d^{-1} \sum_i (\log m_i + Y_i) - .10 \bar{m}^2, \quad \bar{m} = d^{-1} \sum_i m_i.$$

The dense row-normalized matrix B_d is generated once from a recorded seed. Horizon is one, discount is .04, and terminal utility is $\bar{Y} - .03 d^{-1} \sum_i (Y_i - \bar{Y})^2$. The actual state domain is $\mathbb{R}^d$; no artificial lateral stopping condition is imposed. Dense non-linear productivity, shared adjustment cost, common noise, and terminal dispersion couple the coordinates. Neither the solver nor its comparator calls a Riccati or Lyapunov routine, and no exact solution is built into the network.

Strict concavity gives a global Hamiltonian maximizer at each queried state. For $p_i = \partial_i v$, its components satisfy

$$m_i^* = \text{clip} \left((dp_i + .2\bar{m}^*)^{-1}, .02, 2 \right),$$

TABLE VIII

ACTUAL TRAINING IN DENSE NONQUADRATIC CAPITAL MODELS

Method	d	Worst RMS	Worst gap	Failures	Median seconds
NBO	2	0.0161	2.42×10^{-4}	0/3	2.6554
Direct HJB	2	0.0161	0	0/3	3.4870
NBO	5	0.0131	0.0020	0/3	4.1518
Direct HJB	5	0.0129	0	0/3	5.0391
NBO	10	0.0188	0.0052	0/3	6.8652
Direct HJB	10	0.0168	0	0/3	7.6988
NBO	20	0.0320	0.0114	2/3	12.8180
Direct HJB	20	0.0170	0	0/3	13.5001

Maxima are across three seeds, each evaluated at 512 independent points in $[0, 1] \times [-.5, .5]^d$. The failure criterion is RMS residual above .025 or queried action gap above .01 after 600 updates. Both methods use the same critic architecture and global action routine. Times include training diagnostics, not the subsequent Monte Carlo evaluations. These are sampled diagnostics, not uniform economic bounds.

with the upper bound selected when the denominator is nonpositive. The scalar consistency equation for $\bar{m}^*$ is monotone and is solved by bisection. This supplies the same feasible maximization routine to the direct-HJB comparator and to action-gap measurement; it does not supply the value function.

Both networks have two hidden tanh layers of width thirty-two. The critic enforces the terminal payoff exactly. The main design uses three seeds at $d = 2, 5, 10, 20$, 600 Adam updates, and batches of 128 independently sampled state-time points. Exact automatic differentiation supplies the trace. The held-out set contains 512 fixed independent points in $[0, 1] \times [-.5, .5]^d$. A common diagnostic target is residual root mean square at most .025 and maximum queried action gap at most .01. The recorded first crossing is not necessarily sustained; final failures are also reported. The experiment measures real neural approximation and optimization costs rather than exact quadratic identities. It also shows why a diagnostic target is not a welfare certificate. Simulated policy paths can leave the fitted state box; Euler policy-payoff estimates from the initial state need not agree with a critic that has a small sampled residual inside that box. A follow-up expands the fitting and verifi-

cation box to $[-1.5, .5]^d$ and uses 1,000 updates at dimensions ten and twenty. Both designs, paired policy-payoff samples, and forty/eighty-step simulation outputs are retained. Sampling standard errors do not remove time-discretization bias or certify an optimum. Process high-water memory is labeled as such, not interpreted as an isolated per-network allocation. No high-dimensional uniform residual bound is claimed for these sampled quantities.

11.2. *Method-paired payoff evidence and arithmetic replay*

The R7 follow-up evaluates the coupled-capital policies at dimensions ten and twenty, for all three fixed seeds, both fitting domains, and 40, 80, and 160 Euler steps. The same 512 Brownian paths are used by NBO and the direct comparator within every cell. For path j , define $D_j = J_j^{\text{NBO}} - J_j^{\text{direct}}$. We report $\bar{D}$ and the pointwise Student interval $\bar{D} \pm t_{511, .975} s_D / \sqrt{512}$. Pairing removes the irrelevant component of common path variation from the comparison. The complete set of 36 comparisons is reproduced from the pinned raw arrays in the supplement. The old twenty-dimensional fitting box is exited by essentially all recorded paths. Wider-domain fitting eliminates observed exits in the recorded audits and changes both payoff agreement and method differences. Zero observed exits in 512 paths is not a zero exit-probability theorem. Likewise, these intervals quantify pointwise Monte Carlo uncertainty under a specified Euler scheme. They are neither simultaneous confidence statements across all 36 comparisons nor bounds on time bias, optimal value, or the global differential residual. The experiment identifies a consequential coverage failure and its measured correction without redefining the unbounded economic state space as a stopped box.

The one-state consumption certificate has a different evidentiary status. R7 independently replays its previously generated complete leaf cover with `mpmath.iv` at 40 decimal digits, without importing the original interval arithmetic implementation. For the stored seed-11 networks, the NBO policy-loss bound is about .04961 and the direct-HJB bound about .04967. Both meet the stated threshold. This is an *independent interval-arithmetic replay on the previously generated complete cover*, not independent network training, independent cover generation, or a formal machine

TABLE IX
 PAIRED PAYOFF DIFFERENCES AT THE FINEST RECORDED EULER LEVEL

Dim.	Seed	Domain	Difference	95% lower	95% upper
10	11	Narrow	-0.03338	-0.03544	-0.03133
10	11	Wide	0.00052	0.00049	0.00054
10	29	Narrow	-0.03094	-0.03250	-0.02938
10	29	Wide	-0.00003	-0.00004	-0.00002
10	47	Narrow	-0.02755	-0.02941	-0.02569
10	47	Wide	0.00000	-0.00001	0.00001
20	11	Narrow	-0.06302	-0.06572	-0.06033
20	11	Wide	0.00084	0.00080	0.00088
20	29	Narrow	-0.01449	-0.01490	-0.01408
20	29	Wide	0.00409	0.00394	0.00424
20	47	Narrow	-0.20161	-0.20211	-0.20111
20	47	Wide	0.00058	0.00053	0.00062

NBO minus direct payoff on the same 512 paths. Full 40/80/160-step comparisons are in the supplement. Confidence intervals quantify Monte Carlo variation only.

proof. Together with the new complete three-control cover on the two-state preference grid, it demonstrates two different ways of operationalizing verification. Neither removes the dimensional cost of covering an unrestricted high-dimensional continuous state space.

11.3. Exact and stochastic trace computation

Let Σ include any covariance factor and set $B = \Sigma^\top D^2 v \Sigma$. For independent probes with identity covariance,

$$q = \text{tr}(B), \quad \hat{q} = \frac{1}{K} \sum_{k=1}^K z_k^\top B z_k. \quad (52)$$

Gaussian probes have variance $2\|B\|_F^2/K$. If $r = d - q/2$, then

$$\mathbb{E}(d - \hat{q}/2)^2 = r^2 + \frac{1}{4} \text{Var}(\hat{q}). \quad (53)$$

The extra term depends on the critic. Unbiased trace estimation therefore does not give an unbiased squared-residual objective.

Two conditionally independent probe banks repair this particular bias:

$$\hat{L} = (d - \hat{q}^{(1)}/2)(d - \hat{q}^{(2)}/2), \quad \mathbb{E}[\hat{L} \mid t, x, \xi] = r^2. \quad (54)$$

With dominated differentiation, its gradient is also unbiased. The sample product need not be nonnegative and can have substantial variance, so this is an objective identity rather than a guarantee of stable optimization. Exact low-dimensional derivatives remain the independent check. If the actor or deterministic term depends on the same probes, conditional independence must be re-established; the simple identity cannot be reused automatically.

The relevant computational cost depends on the network evaluation and differentiation cost, Brownian rank, covariance factorization, sample count, optimization iterations, and requested accuracy. Forming a dense Hessian involves quadratic storage; Hessian-vector products can avoid it. Neither operation count by itself is a total-complexity theorem. Active control bounds do not universally create gradient discontinuities, and a noisy trace cannot make a correctly projected box action leave its box. Feasibility and value-derivative accuracy are separate diagnostics.

The independent-bank estimator is also exercised during actual training in the coupled ten-state economy. One implementation uses two independent banks of two Gaussian Hessian-vector probes; the matched single-bank ablation squares a four-probe estimate. Exact traces are used only for the common held-out diagnostics. All three seeds, objective signs, optimization histories, and policy-payoff samples are saved. The two training procedures perform similarly at the retained covariance scale; unbiasedness alone does not establish a finite-budget advantage. A separate Monte Carlo test checks both the product objective and its parameter gradient against exact trace expressions with declared tolerances. These calculations validate the estimator’s execution without identifying a random training objective with a deterministic certificate.

12. CONCLUSION

Neural Bellman Operators organize computation around two economic conditions: continuation utility must evaluate the chosen policy, and the policy must satisfy a feasible improvement or unilateral equilibrium condition. The revised formulation preserves recursive utility, constraints, endogenous preference adjustment, temporal selves, and strategic interaction within this common architecture. It also makes terminal data, utility domains, and verification regions explicit.

The main results translate these conditions into value, welfare, and deviation bounds. A positive-weight Bellman approximation provides a viscosity-selection mechanism when classical differentiation is insufficient. The neural representation remains flexible; its economic interpretation comes from the equation and the accompanying error test, not from the printed target or an optimizer’s stationary point.

The computations connect trained multilayer policies to explicit economic error accounts. The stopped-consumption certificate covers the continuous differential domain; the preference-action cover verifies the entire control box on a two-state nodal economy; and the game calculation checks full dynamic unilateral deviations, including deviations from profiles with imperfect fitted one-step equilibria. Non-enumerative continuous actor training, direct control optimization, local-search ablation, gated regression, and projection comparisons disclose both successful policy computations and the cost of the actor. Paired high-dimensional payoff comparisons identify state-coverage failures without claiming an optimality certificate. The continuous economic agenda remains unchanged. Its numerical implementation is sharpened by distinguishing the policy proposed, the computation actually executed, and the domain over which its value or deviation bound has been established.

APPENDIX A: PROOFS

A.1. *Proof of Theorem 2*

Extend the aggregator Lipschitz-continuously outside its invariant interval for this comparison argument. The extension leaves all policy utilities and the candi-

date residual unchanged. For a nonnegative number c , define $d_c(t) = Be^{L(T-t)} + cA_L(T-t)$. Then $d_c(T) = B$, $d_c(t) \geq B$, and $-d'_c = Ld_c + c$. Since a spatially constant shift leaves the generator unchanged and f is L -Lipschitz in utility,

$$F[v + d_c] \geq F[v] - d'_c - Ld_c = F[v] + c, \quad F[v - d_c] \leq F[v] + d'_c + Ld_c = F[v] - c.$$

Because $F[v] = r^a[v] - g^a[v] \in [-e - \eta, e]$, $v + d_{e+\eta}$ is a supersolution and $v - d_{e+\eta}$ is a subsolution. They dominate the required stopping data on every boundary face. A spatially constant shift also preserves an exactly satisfied homogeneous reflected boundary condition. Comparison proves (15).

Apply the same argument to the fixed-policy operator $F^a[v] = -v_t - \mathcal{H}^a[v]$, with $c = e$, to obtain (14). The triangle inequality gives (16), and policy admissibility gives its nonnegative lower bound. The construction uses $L \geq 0$ and $d_c(t) \geq B$ specifically to cover lateral stopping boundaries; it does not incorrectly discount their errors as if every path stopped only at T . For nonsmooth candidates the same argument applies when their residual inequalities are verified in the viscosity sense. The theorem's differential implementation, however, requires the stated smooth candidate. Q.E.D.

A.2. Exact policy improvement

Suppose $v = V^a$ solves the fixed-policy problem and a feasible $\tilde{a}$ satisfies $\mathcal{H}^{\tilde{a}}[v] \geq \mathcal{H}^a[v]$ pointwise. Then $F^{\tilde{a}}[v] \leq 0$ with the same boundary payoff. Policy comparison gives $V^{\tilde{a}} \geq V^a$. This is a policy-space statement. A single gradient step on a domain-averaged actor loss does not automatically meet its pointwise premise. The measured action gap and policy-evaluation errors are retained precisely because approximate neural improvement can violate that premise.

A.3. Proof of Theorem 3

For a fixed state and action, let z and $\tilde{z}$ solve (17) for w and $\tilde{w}$. Strict monotonicity of $z - hf(z)$ gives

$$|z - \tilde{z}| \leq \frac{|P_h^a(w - \tilde{w})|}{1 + \lambda h} \leq q_h \|w - \tilde{w}\|_\infty.$$

The same monotonicity implies order preservation. Taking the maximum over actions preserves both order and the Lipschitz constant. Banach's fixed-point argument supplies the two fixed points on the invariant complete value domain.

Now $\|v - v_h^a\| \leq \|v - T_h^a v\| + q_h \|v - v_h^a\|$, which gives the first bound. Also $\|v - T_h v\| \leq \delta + \epsilon$, giving the second. The triangle inequality gives the third. All norms are over every state of the specified finite model. An omitted action changes T_h and must be accounted for as a separate error, not hidden in δ . *Q.E.D.*

A.4. Proof of Theorem 4

First, (18) gives

$$\|v_h - v_h^*\|_\infty \leq \frac{\delta_h + \epsilon_h}{1 - q_h} = \frac{1 + \lambda h}{\lambda} \frac{\delta_h + \epsilon_h}{h} \rightarrow 0.$$

It remains to identify the limit of the exact monotone scheme. Stability makes the upper and lower half-relaxed limits finite. At a strict local maximum of the upper limit minus a smooth test function, select nearby maximizing grid points. Monotonicity compares the neighboring scheme values to that test function plus the vanishing contact offset. Local consistency then gives the viscosity subsolution inequality. The corresponding minima give the supersolution inequality for the lower limit. Consistent boundary data and the assumed barriers give the required boundary inequalities. Comparison orders the upper limit below the lower limit. The reverse order follows by definition, so the two limits coincide with the unique continuous viscosity solution. The vanishing distance between v_h and v_h^* transfers that convergence to the neural arrays. The discrete policy-value conclusion follows from the third bound. Interpreting a discrete policy as a continuous-time implementation additionally requires convergence of its controlled interpolation and payoff; the theorem does not infer this merely from convergence of its state arrays. This is the usual monotone-scheme selection argument, with the neural approximation error made explicit; see [Barles and Souganidis \(1991\)](#), [Crandall et al. \(1992\)](#). *Q.E.D.*

A.5. Finite-horizon Bellman error accumulation

Let T_n and T_n^a be the time- n Bellman and fixed-policy maps, with Lipschitz constants q_n (which may exceed one). For arrays v_n , define $\delta_n = \|v_n - T_n^a v_{n+1}\|$ and $\epsilon_n = \|T_n v_{n+1} - T_n^a v_{n+1}\|$. If the terminal error is B_N , backward induction gives

$$\|v_n - v_n^*\| \leq \left(\prod_{j=n}^{N-1} q_j \right) B_N + \sum_{k=n}^{N-1} \left(\prod_{j=n}^{k-1} q_j \right) (\delta_k + \epsilon_k).$$

The policy-evaluation bound replaces $\delta_k + \epsilon_k$ by δ_k . Their sum bounds regret. Empty products equal one. For the implicit Lipschitz recursive step, $q_n \leq (1 - hL)^{-1}$; for additive exponential discount, $q_n = e^{-\rho h}$. Thus a per-step error $o(h)$ and consistent terminal data suffice on a fixed finite horizon, without asserting a stationary contraction in an inapplicable utility regime.

A.6. Derivation of the temporal-self condition

A self at (t, x) replaces a^* by a fixed feasible local action a for duration ϵ and then returns to a^* . For each exponential component, the deviation value is

$$\mathbb{E} \left[\int_t^{t+\epsilon} e^{-\rho_j(s-t)} u(a) ds + e^{-\rho_j \epsilon} W_j(t + \epsilon, S_{t+\epsilon}^a) \right].$$

Itô expansion gives its first-order increment relative to $W_j(t, x)$ as $\epsilon[u(a) + (W_j)_t + \mathcal{L}^a W_j - \rho_j W_j] + o(\epsilon)$. Weighting the two components and subtracting their evaluation equations under a^* leaves

$$\epsilon\{u(a) - u(a^*) + (\mathcal{L}^a - \mathcal{L}^{a^*})V\} + o(\epsilon).$$

Nonpositive first-order gain for every feasible action is exactly (45). This derivation assumes the smoothness and local integrability needed for the expansion; it is an equilibrium deviation condition, not an assertion of optimal commitment over a finite interval. The homothetic derivation follows by integrating $e^{-[\rho_j - b(r-m)]s}$ and applying the consumption first-order condition. Q.E.D.

APPENDIX B: REPLICATION AND SCOPE OF NUMERICAL EVIDENCE

Integrated R8 candidate. The present article incorporates the following immutable repository inputs.

R7 source: 5bab4649728860fb655c5371f3705787f10b4a78

R7 evidence: b1409008625f550195ce9ead468391ad2449a240

R7 advisory report: c63f6817f498e1cfe9a653139f7a25bf44a4418a

The preceding main manuscript and supplement are preserved by exact Git blob identity in the R8 archive. R8 source, all executed development configurations, raw arrays, independent certificates, method-paired comparisons, generated tables, and compilation logs belong to the candidate itself. The response answers every B1–B7 and M1–M9 comment of the latest report. The later unexecuted R7 development protocol is not imported as evidence. A table manifest checks each printed number against its input file; the execution manifest identifies the exact R8 source and candidate workflow. Earlier review and revision branches remain unchanged.

The revision directory contains the executable block core, the experiment program, the regression tests, raw numerical outputs, and a response-to-referee document. The reviewed manuscript, supplement, and historical comments are preserved byte-for-byte in the archive, and the review branch remains unchanged. The new main manuscript is authoritative for the revised equations. The archive is not compiled into the revised article and its unsupported numerical statements are not reused as evidence.

The inherited R2 execution contains 29 recorded experiments and eight regression tests, all retained and rerun. The R6 record adds nine primary stopped-consumption runs, six architecture/update sensitivities, six full-domain certificates, six principal NDU neural runs, sixteen NDU reference sensitivities, twenty-four dense-capital primary fits, six stochastic-trace fits, twelve coverage follow-ups, and six principal dynamic-game solves. Pilot failures and coarse-game counterchecks are retained separately rather than counted as successful primary runs. The original referee diagnostics continue to concern their pinned original sources.

The initial development summary and the corresponding script are preserved separately. Inspection found a sign error in the *reported* quadratic residual formula

even though the Lyapunov solutions agreed with Riccati values. The formula was corrected, its acceptance test strengthened to include that residual, and the complete experiment suite rerun. The wealth stopping boundary was also imposed explicitly at all NDU time levels before the final run. The earlier results are marked superseded diagnostics rather than silently relabelled as final evidence.

The R6 directory contains complete source, generated tables, model and numerical arrays, optimizer histories, certificate covers, tests, and a provenance manifest. The replication workflow materializes the revision source, independently reruns its computations, and commits the resulting evidence only to the designated R6 candidate branch; an exact-source snapshot and raw-result hashes identify each execution. Local execution, remote replication, and independent referee acceptance remain different facts. No historical GPU benchmark, unexecuted deep-BSDE comparison, unrestricted neural Cournot equilibrium, or empirical identification result is inferred from the computations reported here.

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